

3D MODEL · SPEC CONVERSION

E3D → PCF (CAESAR II)

Extracts E3D model to CAESAR II PCF input. Single flat MAC engine for TTY + Tkinter login GUI.

Key Purpose

Why this program

1. Turn the live E3D model into CAESAR II input directly

- Selected pipes are written straight to PCF files — no manual re-modelling in the stress tool
- **Effect:** The stress engineer gets a ready-to-import model in minutes with zero transcription error

2. Runs inside the E3D DESIGN session as one flat MAC engine

- No PML form install or add-on registration — a single inlined TTY macro drives the console
- **Effect:** Portable one-file tool that works on any project login and can later compile to a single exe

3. Batch whole systems, not one pipe at a time

- Collect every PIPE under CE, or pull complete systems from the project ISO .lis list
- **Effect:** A full line or unit is exported in one pass instead of picking components by hand

4. Clean the model as it exports

- Drop small-bore noise, add insulation density, and set the CAESAR II vertical axis (Y or Z up)
- **Effect:** The imported model is scoped and oriented correctly, with no axis flip or trivial-bore clutter

Required inputs : Logged-in E3D DESIGN session (project / MDB) + PIPE(s) selected from CE or the ISO .lis system list

Main Window – Item Guide

The screenshot shows the CAESAR II Interface (PCF) window. The interface is divided into several sections:

- E3D Setup / LOGIN**: Contains buttons for "Select **1** BAT" and "Select **2** S BAT". Below these is a green button labeled "LOGIN (Open E**3** DESIGN Console)".
- Login Info**: A form with fields for "User:" (SYSTEM), "PROJECT:" (empty), "Password:" (*****), "MDB:" (PIPE), "Module:" (DESIGN), and "Device:" (GRAPHICS). A red circle **4** is around the "Module:" dropdown.
- Buttons**: Two buttons are present: "Add P**5** om CE" and "Add Syst**6** m Iso List".
- Element**: A list box for selecting elements, with a red circle **7** around it.
- PCF Files (latest first)**: A list box for PCF files, with a red circle **8** around it.
- Exclude Bore Size <=**: A text box with the value "25", with a red circle **9** around it.
- Insulation Density**: A text box with the value "200".
- CaesarII Orientation**: Radio buttons for "Y is **10**" and "Z is U".
- Buttons**: Four buttons: "Rem Selected List", "Clear All **11**", "Add Selected", and "Rem Selected".
- Select **12** Folder**: A button to select an output folder.
- Output**: A text box showing "Output: default (\$PROJ ISO\PCF\)".

1. Select E3D BAT (project launcher batch)
2. Select EVARS BAT (environment variables batch)
3. LOGIN — open E3D DESIGN console session
4. Login Info: User / Project / Password / MDB / Module
5. Add Pipe from CE (collect PIPEs under current element)
6. Add System from Iso List (project *.lis systems)
7. Element list — selected pipes / systems to export
8. PCF Files (generated output, latest first)
9. Exclude Bore Size <= / Insulation Density
10. CaesarII Orientation (Y is U / Z is U vertical axis)
11. Add / Rem Selected · Rem List · Clear All
12. Select Output Folder (default \$PROJ ISO\PCF\)

Overview

What it does

1. Extracts E3D model to CAESAR II PCF input
2. Single flat MAC engine for TTY
3. Tkinter login GUI

Category: 3D Model · Spec Conversion

Folder: CAESARII

Main scripts: e3dtopcf.py

Scripts / Status: 5 scripts · Operational

Tags: E3D · PCF · CAESAR II · Stress

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: e3dtopcf.py.
- Output samples and the program window above reflect the current build.

Thank You

E3D → PCF (CAESAR II) · Plant Engineering Total Service System